

50.021 – Artificial Intelligence

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Week 10 Theory Homework - Planning

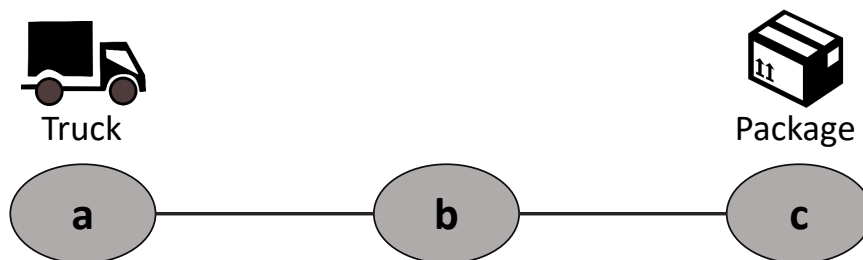
[The following notes are compiled from various sources such as textbooks, lecture materials, Web resources and are shared for academic purposes only, intended for use by students registered for a specific course. In the interest of brevity, every source is not cited. The compiler of these notes gratefully acknowledges all such sources.]

Due: 11th Apr 2024, 11:59pm

Submission: via eDimension

1 Logistic Problem I

Consider the following logistic problem. There are three locations a , b and c , with a truck at a and package at c . The truck is able perform the following actions: (i) $\text{move}(x,y)$: move from location x to y ; (ii) $\text{load}(x)$: load a package at location x ; and (iii) $\text{unload}(x)$: unload the package at location x . The truck can only move between adjacent locations, e.g., a to b , b to c (You can assume that these static facts are already modelled/defined).



Given the start state in the above diagram, your goal is to get the package to location b . Formulate this logistic problem using the STRIPS representation and answer the following:

- List down the propositional variables (facts).
- Specify the operators (actions), including the pre-conditions and post-conditions.
- Specify the initial state.
- List down the goal state/specification.

2 Logistic Problem II

Based on your STRIPS formulation from Q1 (Logistic Problem I), answer the following:

- What is the optimal solution to this problem?
- Make this a delete-relaxed problem. What are the changes to the original STRIPS formulation you made?
- Based on this delete-related problem, list down all the facts F_x and actions A_x at levels $x = \{0, 1, \dots, M\}$.

3 Logistic Problem III

Based on your answer from Q2 (Logistic Problem II), answer the following:

- What is the optimal solution to this delete-relaxed problem? What is this heuristic called?
- What is the value of h_{add} ? Explain why.
- What is the value of h_{max} ? Explain why.

4 Generic Planning I

Consider a STRIPS problem with propositional variables (facts) m, n, o, p , and the below STRIPS actions with their pre/post-conditions.

Action	Pre	Add	Del
A	m	n,o	$\emptyset$
B	m,o	p	m
C	p	m	p
D	n,o	p	o

Given an initial state $s = \{m\}$ and goal specification $g = \{m, n, o, p\}$, answer the following questions:

- What is the value of h_+ ? Explain why.
- What is the value of h_{add} ? Explain why.
- What is the value of h_{max} ? Explain why.

5 Generic Planning II

Based on the same STRIPS formulation in Q4 (Generic Planning I). Now, based on initial state $s = \{p\}$ and goal specification $g = \{m, n, o, p\}$, answer the following questions:

- What is the value of h_+ (if any)? Explain why.
- What is the value of h_{add} (if any)? Explain why.
- What is the value of h_{max} (if any)? Explain why.

1. a) $P: \text{truckAt}(x)$
 $\text{packageAt}(x)$

b) $O:$

$\text{Move}(x, y)$
pre - $\text{truckAt}(x), \neg \text{truckAt}(y)$
post - $\neg \text{truckAt}(x), \text{truckAt}(y)$

$\text{Load}(x)$
pre - $\text{packageAt}(x), \text{truckAt}(x)$
post - $\neg \text{packageAt}(x), \text{truckAt}(x)$

$\text{Unload}(x)$
pre - $\neg \text{packageAt}(x), \text{truckAt}(x)$
post - $\text{packageAt}(x), \text{truckAt}(x)$

c) $I:$

$\text{truckAt}(a)$
 $\text{packageAt}(c)$

d) $G:$

$\text{truckAt}(b)$
 $\text{packageAt}(b)$

2. a) Optimal solution: Truck goes from a to c, picks up the package, goes back to b and unloads the package

b) The post conditions for the actions are changed, except for $\text{unload}(x)$

$\text{Move}(x, y): \text{truckAt}(y)$
 $\text{load}(x): \text{truckAt}(x)$

$$c) F_0 = \text{truckAt}(A), \text{packageAt}(C)$$

$$A_1 = \text{Move}(A, B)$$

$$F_1 = \text{truckAt}(B), \text{packageAt}(C)$$

$$A_2 = \text{Move}(B, C)$$

$$F_2 = \text{truckAt}(C), \text{packageAt}(C)$$

$$A_3 = \text{Load}(C)$$

$$F_3 = \text{truckAt}(C)$$

$$A_4 = \text{Move}(C, B)$$

$$F_4 = \text{truckAt}(B), \neg \text{packageAt}(B)$$

$$A_5 = \text{Unload}(B)$$

$$F_5 = \text{truckAt}(B), \text{packageAt}(B)$$

$$3. a) h^+ \text{ heuristic} = 4$$

$$b) h_{add} = 1 + 5 = 6$$

$$c) h_{max} = 5$$

4. a) $h_+ = 2$

b) $h_{add} = 1+2=3$

c) $h_{max} = 2$

A: $pre = m$

$post = m, n, 0$

B: $pre = m, n, 0$

$post = m, n, 0, p$

5. a) $h_+ = 2$

b) $h_{add} = 1+2$
 $= 3$

c) $h_{max} = 2$

C: $pre = p$

$post = p, m$

A: $pre = p, m$

$post = n, m, 0, p$